

Executive Summary

This material covers fundamental concepts in cellular biology, including the structure and function of cells, photosynthesis, cellular respiration, and genetic processes. It also explores cell transport mechanisms, enzyme function, homeostasis, and key human body systems.

KEY TAKEAWAYS

The cell is the basic structural and functional unit of all living organisms, as defined by the cell theory.

Photosynthesis transforms light energy into chemical energy using the equation $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

Cellular respiration involves glycolysis, the Krebs cycle, and the electron transport chain, with the summary equation $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$.

DNA carries genetic instructions and undergoes replication, transcription, and translation to produce proteins.

Homeostasis maintains stable internal conditions, regulated by feedback mechanisms such as insulin and glucagon for blood glucose control.

Topics

TOPIC 1

Cellular Biology

Defines the cell as the basic unit of life and outlines its key components and functions.

Cell Theory

The cell theory states that all living things are composed of cells.

New cells are produced from existing cells.

Cell Structure

The nucleus is the membrane-bound organelle that contains genetic material.

Mitochondria produce ATP through cellular respiration.

Organisms

Every organism is composed of one or more cells.

TOPIC 2

Photosynthesis

Describes the process by which green plants convert light energy into chemical energy.

Process Overview

Photosynthesis is the process by which green plants transform light energy into chemical energy.

Location

Photosynthesis occurs in the chloroplast, which contains the green pigment chlorophyll.

Key Components

Chlorophyll is the pigment that absorbs light energy.

Chemical Equation

The summary equation of photosynthesis is $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

Stages

Light-dependent reactions occur in the thylakoid membranes and produce ATP and NADPH.

The Calvin cycle is the light-independent stage that fixes carbon dioxide into glucose.

TOPIC 3

Cellular Respiration

Explains the three main stages of cellular respiration and its role in energy production.

Stages

Glycolysis occurs in the cytoplasm and breaks down glucose into two molecules of pyruvate.

The Krebs cycle occurs in the mitochondrial matrix.

The electron transport chain is the final stage and occurs on the inner mitochondrial membrane.

Anaerobic Process

Fermentation is the anaerobic process that regenerates NAD^+ when oxygen is unavailable.

Chemical Equation

The summary equation of cellular respiration is $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$.

TOPIC 4

Cell Transport

Describes mechanisms by which substances move across the cell membrane.

Cell Membrane

The cell membrane is the selectively permeable barrier that controls what enters and leaves the cell.

Passive Transport

Diffusion is the passive movement of molecules from an area of high concentration to low concentration.

Osmosis

Osmosis is the diffusion of water across a selectively permeable membrane.

Active Transport

Active transport is the movement of substances against their concentration gradient, requiring energy in the form of ATP.

Bulk Transport

Endocytosis is the process by which the cell engulfs large particles by wrapping the membrane around them.

TOPIC 5

DNA and Protein Synthesis

Explains the flow of genetic information from DNA to proteins.

DNA

DNA is the molecule that carries genetic instructions for the development and functioning of living organisms.

DNA Replication

DNA replication is the process by which a double-stranded DNA molecule is copied to produce two identical DNA molecules.

Transcription

Transcription is the process by which the DNA sequence of a gene is copied into messenger RNA.

Translation

Translation is the process by which the ribosome builds a protein from the mRNA sequence.

Genetic Code

Each three-nucleotide codon on mRNA specifies one amino acid.

The genetic code is the set of rules by which information encoded in genetic material is translated into proteins.

TOPIC 6

Cell Division

Describes the process of mitosis and the cell cycle.

Mitosis

Mitosis is the process of nuclear division that produces two genetically identical daughter cells.

Cell Cycle

The cell cycle consists of interphase, mitosis, and cytokinesis.

Interphase

Interphase is the stage during which the cell grows, copies its DNA, and prepares for division.

Mitotic Stages

Prophase is the first stage of mitosis, during which chromosomes condense.

Anaphase is the stage during which sister chromatids are pulled apart to opposite poles of the cell.

TOPIC 7

Genetics

Covers the basics of heredity, genes, and genetic inheritance.

Gene and Allele

A gene is the basic unit of heredity, a segment of DNA that codes for a specific trait.

An allele is an alternative form of a gene at the same locus.

Genotype and Phenotype

The genotype is the genetic makeup of an organism.

The phenotype is its observable characteristics.

Allele Expression

A dominant allele is expressed when only one copy is present.

A recessive allele is only expressed when two copies are present.

Punnett Squares

Punnett squares are used to predict the probability of offspring inheriting particular traits.

TOPIC 8

Enzymes

Explains the role of enzymes as biological catalysts and their function.

Enzyme Definition

An enzyme is a protein that acts as a biological catalyst, speeding up chemical reactions without being consumed.

Active Site

The active site is the region of an enzyme where the substrate binds.

Enzyme Function

The induced fit model describes how the enzyme changes shape to accommodate the substrate.

Optimal Conditions

Optimal temperature is the temperature at which an enzyme works fastest.

Denaturation

Denaturation is the loss of an enzyme's three-dimensional shape, which destroys its function.

TOPIC 9

Homeostasis

Describes the maintenance of stable internal conditions despite external changes.

Definition

Homeostasis is the maintenance of a stable internal environment despite external changes.

Feedback Mechanisms

Negative feedback reverses a change, bringing the system back to its set point.

Positive feedback amplifies a change and drives the system further from its set point.

Examples

The normal human body temperature is maintained at approximately 37 degrees Celsius.

Blood glucose concentration is regulated by the hormones insulin and glucagon.

Insulin is secreted when blood glucose is high, promoting glucose uptake.

Human Systems

Provides an overview of key human body systems and their functions.

Circulatory System

The circulatory system transports oxygen, nutrients, and waste products throughout the body.

The heart is the muscular organ that pumps blood through the circulatory system.

Respiratory System

The respiratory system is responsible for gas exchange, bringing oxygen into the body and removing carbon dioxide.

Nervous System

The nervous system coordinates rapid responses through electrical signals called nerve impulses.

Endocrine System

The endocrine system regulates long-term processes through chemical messengers called hormones.

Terms & Definitions

Term	Definition
Allele	An allele is an alternative form of a gene at the same locus.
ATP	The mitochondria is the organelle responsible for producing ATP through cellular respiration.
Active Transport	Active transport is the movement of substances against their concentration gradient, requiring energy in the form of ATP.
Anaphase	Anaphase is the stage during which sister chromatids are pulled apart to opposite poles of the cell.
Cell	The cell is defined as the basic structural and functional unit of all living organisms.
Cell Division	Mitosis is the process of nuclear division that produces two genetically identical daughter cells.
Cell Membrane	The cell membrane is the selectively permeable barrier that controls what enters and leaves the cell.
Cell Respiration	Cellular respiration consists of three main stages.
Chlorophyll	Photosynthesis occurs in the chloroplast, which contains the green pigment chlorophyll.
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Cytokinesis	The cell cycle consists of interphase, mitosis, and cytokinesis.
Denaturation	Denaturation is defined as the loss of an enzyme's three-dimensional shape, which destroys its function.
Diffusion	Diffusion is the passive movement of molecules from an area of high concentration to low concentration.
DNA	DNA is defined as the molecule that carries the genetic instructions for the development and functioning of living organisms.
DNA Replication	DNA replication is the process by which a double-stranded DNA molecule is copied to produce two identical DNA molecules.
Endocytosis	Endocytosis is the process by which the cell engulfs large particles by wrapping the membrane around them.
Enzyme	An enzyme is defined as a protein that acts as a biological catalyst, speeding up chemical reactions without being consumed.
Gene	A gene is defined as the basic unit of heredity, a segment of DNA that codes for a specific trait.
Genotype	The genotype is the genetic makeup of an organism, while the phenotype is its observable characteristics.
Glycolysis	Glycolysis occurs in the cytoplasm and is the first stage, during which

Term	Definition
	glucose is broken down into two molecules of pyruvate.
Homeostasis	Homeostasis is defined as the maintenance of a stable internal environment despite external changes.
Interphase	Interphase is the stage during which the cell grows, copies its DNA, and prepares for division.
Krebs Cycle	The Krebs cycle occurs in the mitochondrial matrix.
Mitochondria	The mitochondria is the organelle responsible for producing ATP through cellular respiration.
Mitosis	Mitosis is the process of nuclear division that produces two genetically identical daughter cells.
NADPH	The light-dependent reactions occur in the thylakoid membranes and produce ATP and NADPH.
Nucleus	The nucleus is the membrane-bound organelle that contains the genetic material.
Osmosis	Osmosis is defined as the diffusion of water across a selectively permeable membrane.
Phenotype	The genotype is the genetic makeup of an organism, while the phenotype is its observable characteristics.
Photosynthesis	Photosynthesis is defined as the process by which green plants transform light energy into chemical energy.
Prophase	Prophase is the first stage of mitosis, during which chromosomes condense.
Protein Synthesis	Translation is defined as the process by which the ribosome builds a protein from the mRNA sequence.
Punnett Squares	Punnett squares are used to predict the probability of offspring inheriting particular traits.
RNA	Transcription is the process by which the DNA sequence of a gene is copied into messenger RNA.
Transcription	Transcription is the process by which the DNA sequence of a gene is copied into messenger RNA.
Translation	Translation is defined as the process by which the ribosome builds a protein from the mRNA sequence.

Key Facts & Formulas

Formula / Fact	Context
respiration is $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$. $\text{C}_6\text{H}_{12}\text{O}_6$	respiration is $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$. oduce glucose and release oxygen. The summary equation of photosynthesis is $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. The light-dependent reactions occur in the thylakoid membranes and produce ATP and